

Proof. Let $a, b \geq 0$ and let $p, q > 1$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$.

Degenerate cases. If $a = 0$ or $b = 0$ then $ab = 0$ while the right-hand side $\frac{a^p}{p} + \frac{b^q}{q}$ is non-negative, so the inequality holds. Hence we may assume

$$a > 0, \quad b > 0.$$

A convex function. Define $\phi(x) = \frac{x^p}{p}$ for $x \geq 0$. Then

$$\phi'(x) = x^{p-1}, \quad \phi''(x) = (p-1)x^{p-2} > 0 \quad (p > 1, x > 0),$$

so ϕ is convex on $[0, \infty)$. For a convex differentiable function the supporting-line inequality holds:

$$\phi(a) \geq \phi(x_0) + \phi'(x_0)(a - x_0) \quad (\forall a \geq 0, x_0 > 0). \quad (1)$$

Choice of the tangency point. Take $x_0 = b^{q-1}$. Since the exponents are conjugate,

$$q = \frac{p}{p-1}, \quad q-1 = \frac{1}{p-1},$$

and therefore $(q-1)(p-1) = 1$. Consequently

$$\phi'(x_0) = (b^{q-1})^{p-1} = b^{(q-1)(p-1)} = b, \quad (2)$$

$$\phi(x_0) = \frac{(b^{q-1})^p}{p} = \frac{b^{p(q-1)}}{p} = \frac{b^q}{p}, \quad (3)$$

because $p(q-1) = q$.

Applying (1). Insert $x_0 = b^{q-1}$ together with (2) and (3) into (1):

$$\frac{a^p}{p} \geq \frac{b^q}{p} + b(a - b^{q-1}).$$

Expanding the right-hand side yields

$$\frac{a^p}{p} \geq \frac{b^q}{p} + ab - b^q.$$

Rearranging,

$$\frac{a^p}{p} + \left(1 - \frac{1}{p}\right)b^q \geq ab.$$

Since $1 - \frac{1}{p} = \frac{p-1}{p} = \frac{1}{q}$ (by $\frac{1}{p} + \frac{1}{q} = 1$), we obtain

$$\frac{a^p}{p} + \frac{b^q}{q} \geq ab. \quad (4)$$

Thus Young's inequality holds for all $a, b > 0$.

Equality condition. In (1) equality occurs iff $a = x_0$. With our choice of x_0 this means $a = b^{q-1}$. Raising both sides to the power p and using $p(q-1) = q$ gives $a^p = b^q$. Conversely, if $a^p = b^q$ then $a = b^{q-1}$ and each step above is an equality. The trivial case $a = b = 0$ also gives equality. Hence

$$ab = \frac{a^p}{p} + \frac{b^q}{q} \iff a^p = b^q.$$

Conclusion. Combining the zero case with the argument for $a, b > 0$ we have proved Young's inequality

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q},$$

with equality precisely when $a^p = b^q$ (including $a = b = 0$). $\square$

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